

Ganesh Gupta

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EXPERIENCE

Oracle | *Software Engineer, AI Platform*

Jun 2025 – Present

- Led a production **agentic system** on **OpenAI Agents SDK**, **LangGraph**, and **MCP** tool use that plans multi-step tool calls over enterprise data, handling **millions of daily transactions** at **99%+ uptime**.
- Built distributed **search and retrieval** infrastructure on **Elasticsearch** with **knowledge graph** enrichment, served through typed **REST APIs**.
- Implemented **GraphRAG** in production, improving grounding and accuracy over standard vector **RAG** on complex **multi-hop** questions.
- Built **evaluation harnesses** and **regression suites** that benchmark agent versions on hallucination, **grounding**, and safety, gating every release on measured results.

QureAI | *ML Infrastructure Engineer*

Mar 2026 – Present

- Owned **retrieval relevance** for a production product, raising **top-3 recall from 61% to 89%** on a 2K-query eval set with **hybrid dense-sparse retrieval** and **re-ranking**.
- Designed a **Neo4j knowledge graph** of domain entities and relationships, deployed to production for **graph-aware retrieval** in a regulated healthcare product.
- Built **PySpark** jobs and **Airflow** DAGs for large-scale **feature engineering** and dataset preparation feeding model training and evaluation.

Nomura Research Institute Financial Technologies | *Software Engineer*

Sep 2019 – Aug 2023

- Architected and operated **25+ production Python and Java microservices** for **1M+ users** in **financial services** at **99.9% uptime** on **AWS**.
- Cut nightly batch aggregation from **8 minutes to 90 seconds** by optimizing **SQL and PL/SQL** on a financial RDBMS with indexed views and partitioning.
- Cut production escalations **60%** in one quarter by standardizing API contracts and failure handling, driving **root cause analysis** and post-incident reviews.
- Led **architecture and code reviews** and mentored engineers on system design, reliability, and secure API practices.

PROJECTS

LangFetch

tinyurl.com/LangFetch

- **LangGraph** agent that decomposes natural-language **financial queries** into dependency-ordered **SQL** subqueries with **Pinecone** retrieval, served via **FastAPI** as an **MCP** server.

RootCause AI

github.com/ganeshhgupta/rootcause-ai

- **LangGraph** multi-agent system that routes production alerts to specialized sub-agents via **tool-use** chains and writes structured root-cause reports, cutting **MTTR 65%**.

Watch-your-LLM

github.com/ganeshhgupta/watch-your-llm

- **LLM evaluation** platform with rubric-based scoring, **hallucination detection**, regression alerting, a **FastAPI** backend, and a **React/TypeScript** dashboard.

EDUCATION

University of Texas at Arlington | *M.S. Computer Science, GPA 4.0/4.0*

Aug 2023 – May 2025

MCKV Institute of Engineering | *B.Tech Computer Science*

Aug 2016 – Jul 2020

SKILLS

Languages: Python, TypeScript, Java, SQL

Search & Retrieval: Hybrid Retrieval, Re-ranking, Embeddings, Elasticsearch, Pinecone, FAISS, GraphRAG, Neo4j

Agents & Evals: MCP, OpenAI Agents SDK, LangGraph, Tool Calling, Relevance Evals, A/B Testing, Regression Suites

Data Infra: PySpark, Airflow, Kafka, PostgreSQL, Redis, AWS, GCP, Kubernetes

Certifications: AWS Certified Developer Associate, GCP Professional MLOps Engineer